

University of Asia Pacific

Department of Computer Science and Engineering

CSE 438: Robotics Lab

LAB REPORT

Experiment Number: 01

Experiment Title: Designing of a Line Following Robot with IR Sensors

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1. Experiment Name

Designing and Implementing a Line-Following Robot Using Arduino UNO

2. Objective

The main objective of this experiment is to design and implement an autonomous line-following robot using an **Arduino UNO**, **IR sensors**, and **DC gear motors**.

The robot will detect a predefined line on the ground using two IR sensors and automatically control the direction of its motors to follow the line.

Specific Objectives

- To interface IR sensors with Arduino UNO.
- To control DC motors using an L293D motor driver.
- To detect the position of a line using two IR sensors.
- To make the robot move forward, turn left, and turn right automatically.
- To understand the basic working principle of an autonomous mobile robot.

3. Apparatus / Hardware & Software Requirements

- List all required tools and components:

Hardware Requirements:

Component	Quantity	Purpose
Arduino UNO	1	Main microcontroller
IR Sensor Module	2	Detect the line
L293D Motor Driver IC	1	Control the DC motors
DC Gear Motor	2	Drive the robot
9V Battery	1	Power source
Breadboard	1	Circuit connection
Jumper Wires	Several	Electrical connections
5V Voltage Regulator	1	Provide regulated 5V supply
USB Cable	1	Programming Arduino

Software Requirements

- Arduino IDE
- Tinkercad Circuits (for simulation/design)
- Arduino programming language (C/C++)

4. Circuit Diagram / Schematic

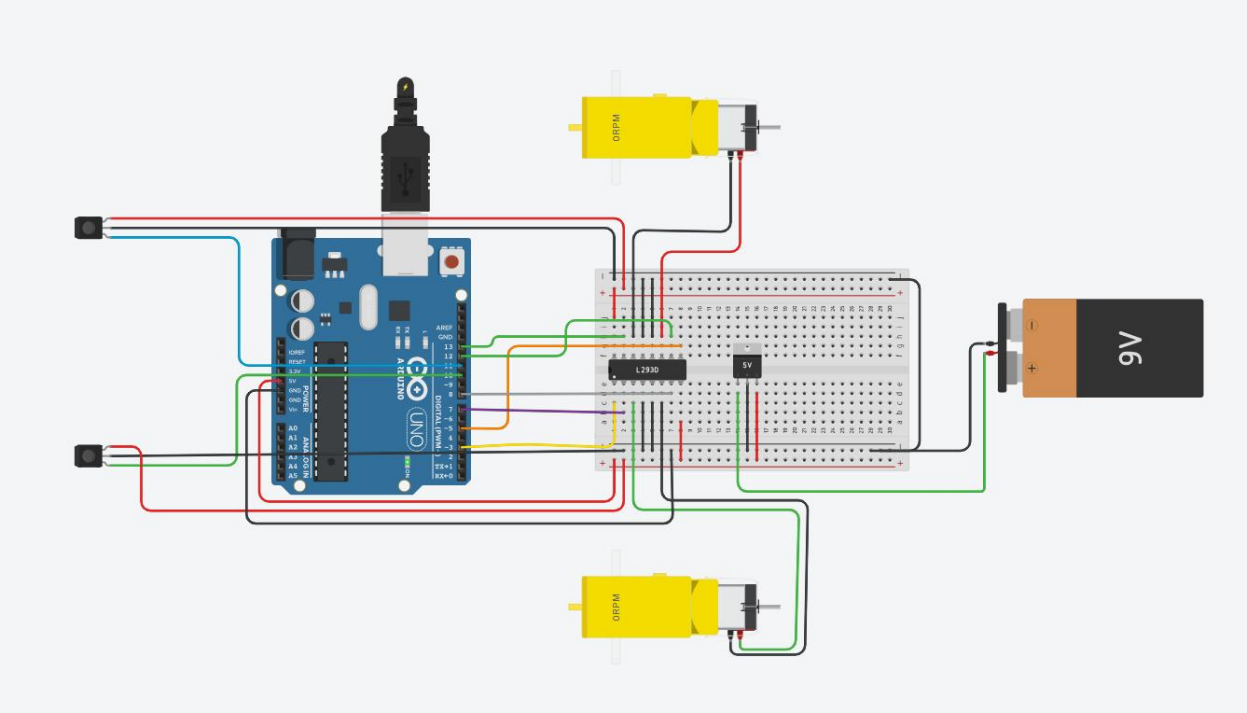


Fig: Line-following robot

5. Code / Assembly Program

```
// C++ code
//
int i = 0;
int j = 0;
void setup()
{
  pinMode(10, INPUT);
  pinMode(11, INPUT);
  pinMode(12, OUTPUT);
  pinMode(13, OUTPUT);
  pinMode(7, OUTPUT);
  pinMode(8, OUTPUT);
  pinMode(10, OUTPUT);
  pinMode(11, OUTPUT);
}
void loop()
{
  // Robot moves forward
  if (digitalRead(10) == HIGH && digitalRead(11) == HIGH) {
    digitalWrite(12, HIGH);
    digitalWrite(13, LOW);
    digitalWrite(7, HIGH);
    digitalWrite(8, LOW);
  }
  // Robot turns right
```

```
if (digitalRead(10) == HIGH && digitalRead(11) == LOW) {
  digitalWrite(12, LOW);
  digitalWrite(13, LOW);
  digitalWrite(7, HIGH);
  digitalWrite(8, LOW);
}
// Robot turns left
if (digitalRead(10) == LOW && digitalRead(11) == HIGH) {
  digitalWrite(12, HIGH);
  digitalWrite(13, LOW);
  digitalWrite(7, LOW);
  digitalWrite(8, LOW);
}
// Robot stops and turns around
if (digitalRead(10) == LOW && digitalRead(11) == LOW) {
  digitalWrite(12, LOW);
  digitalWrite(13, LOW);
  digitalWrite(7, LOW);
  digitalWrite(8, LOW);
  delay(1000); // Wait for 1000 millisecond(s)
  digitalWrite(10, HIGH);
  digitalWrite(11, LOW);
  delay(5000); // Wait for 5000 millisecond(s)
  digitalWrite(12, LOW);
  digitalWrite(13, LOW);
  digitalWrite(7, LOW);
```

```
    digitalWrite(8, LOW);
    delay(1000); // Wait for 1000 millisecond(s)
  }
}
```

6. Output / Observations

After uploading the program to the Arduino UNO, the robot responds according to the signals received from the two IR sensors.

Observed Behavior

Left Sensor	Right Sensor	Robot Action
LOW	LOW	Moves Forward
LOW	HIGH	Turns Left
HIGH	LOW	Turns Right
HIGH	HIGH	Stops

The IR sensors continuously detect the line and send digital signals to the Arduino. Based on these signals, the Arduino controls the motors through the L293D motor driver.

When the robot moves away from the line, one of the sensors detects the change and the corresponding motor is controlled to bring the robot back toward the line.

7. Result

The line-following robot was successfully designed and implemented using an **Arduino UNO, two IR sensors, an L293D motor driver, and two DC gear motors**.

The robot was able to detect the line using the IR sensors and change its movement according to the sensor readings.

Therefore, the objective of the experiment was successfully achieved.

8. Conclusion

In this experiment, we developed a simple line-following robot using an Arduino UNO. Through this project, we learned how to connect and use IR sensors with a microcontroller and how to control DC motors using the L293D motor driver.

We also learned how the Arduino can read sensor signals and make decisions to move the robot forward, turn left or right, and stop when needed. Overall, this experiment gave us practical experience with Arduino programming, IR sensors, motor control, and the basic working principles of an autonomous robot.